



Carnegie Mellon University

Master of
Software Engineering

Module views, module decomposition view

LG Architecture Training Program

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Agenda

➔ Module views

View showing module decomposition

View showing module use

Layered style

MVC style

Data model/domain model

Hexagonal architecture

Microkernel pattern

Module views

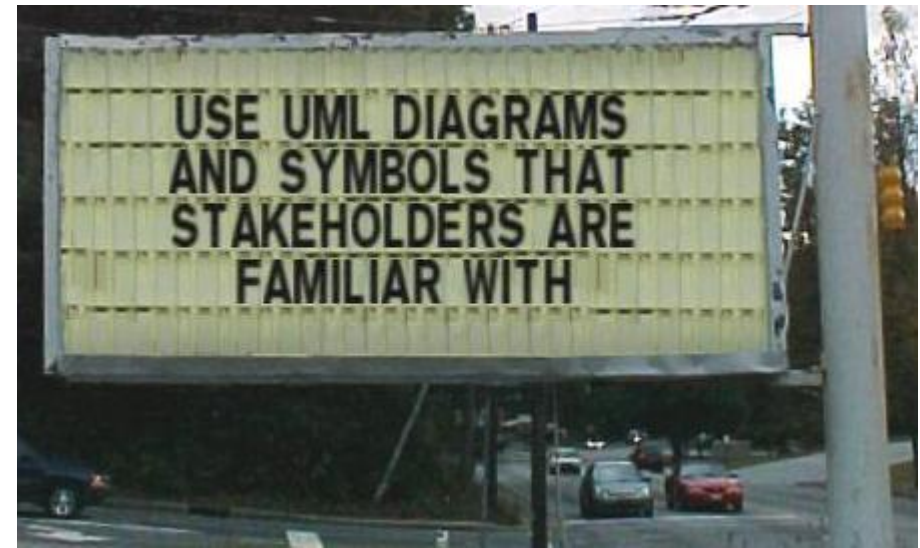
- Show structures of the system in terms of units of implementation
- **Elements**: modules, i.e., implementation units
- **Relations**:
 - **A is part of B**: part-whole relation among modules
 - **A depends on B**: dependency relation among modules
 - **A is a B**: specialization/generalization relation among modules, or interface realization



The Unified Modeling Language



- UML is a **visual** language to specify, design, and document artifacts of a software system
- It is a de facto and de jure standard (OMG) for software design representation
- UML has 14 types of diagrams



Example of prejudice towards UML...



[Youtube video](#), minute 12', and then ... QA minute 29:40

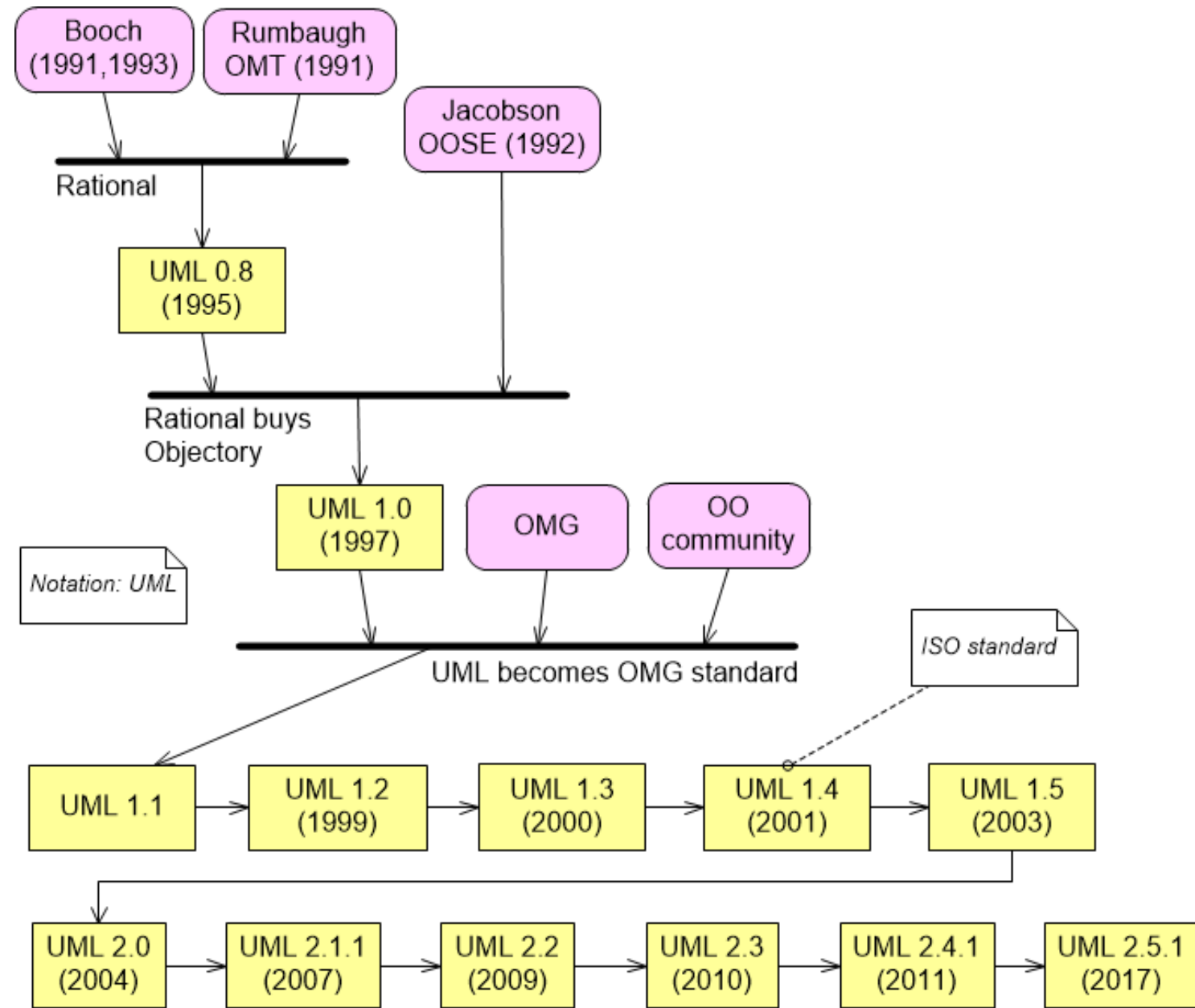
UML history



Grady
Booch

Ivar
Jacobson

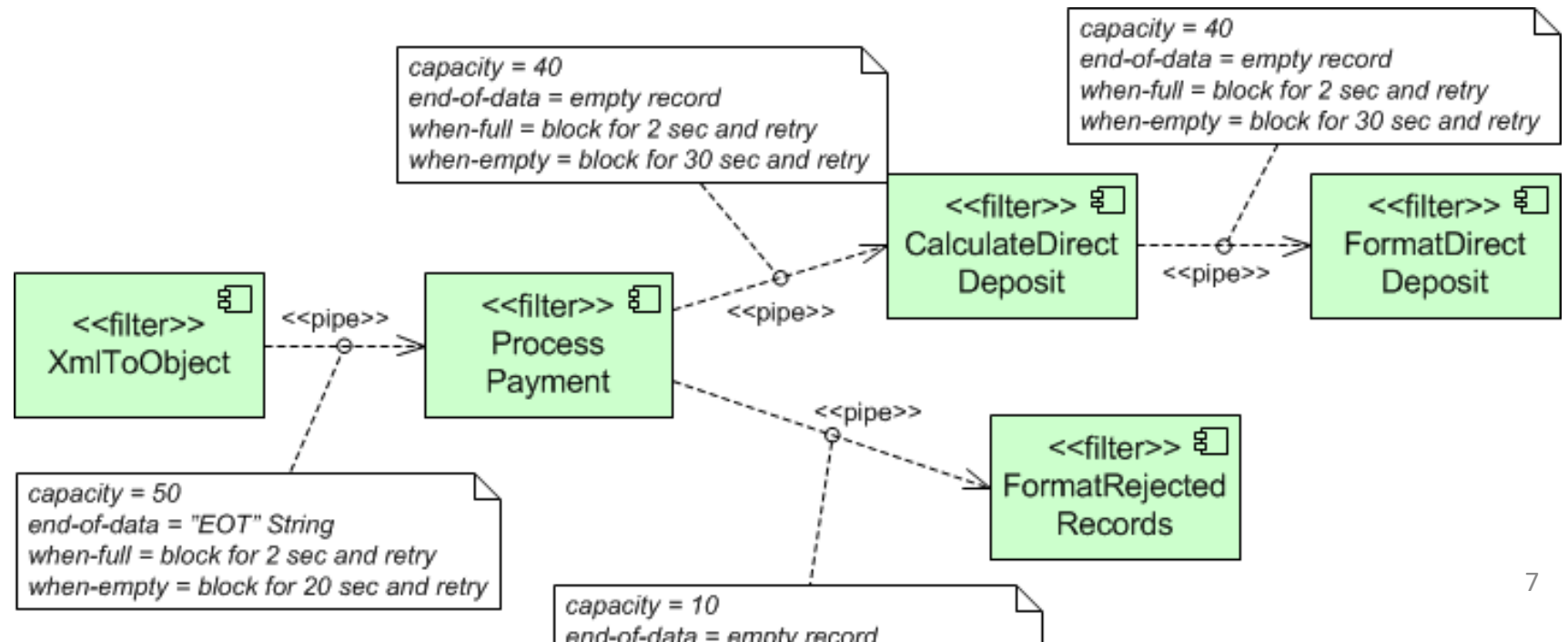
Jim
Rumbaugh



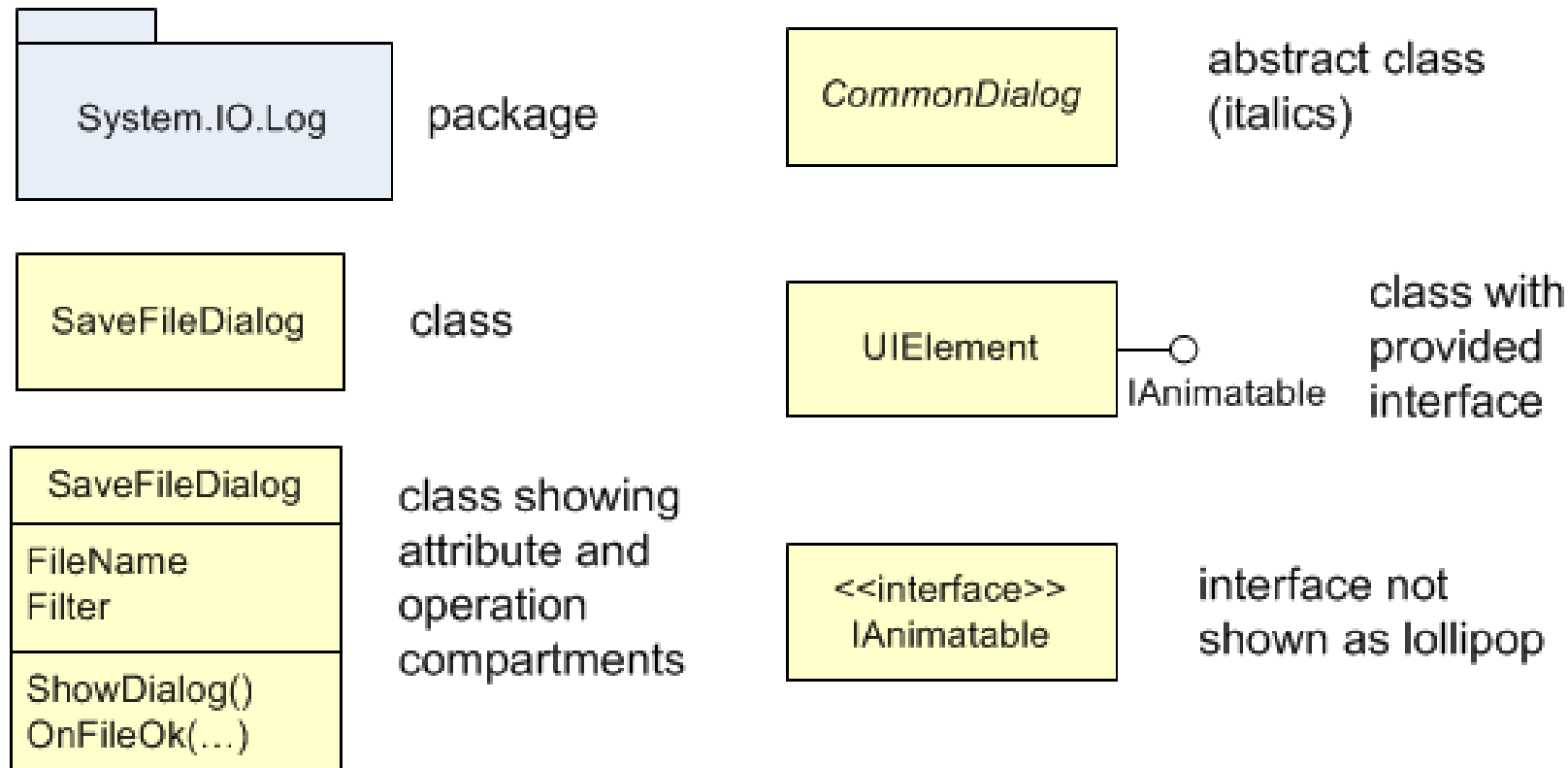
UML extension mechanisms

- **Stereotype**: name in “<< >>” that customizes or specializes a type of element or relation
- **Tagged value**: tag string (key) and value pair that is associated to a stereotyped element or relation

There are also constraints and profiles



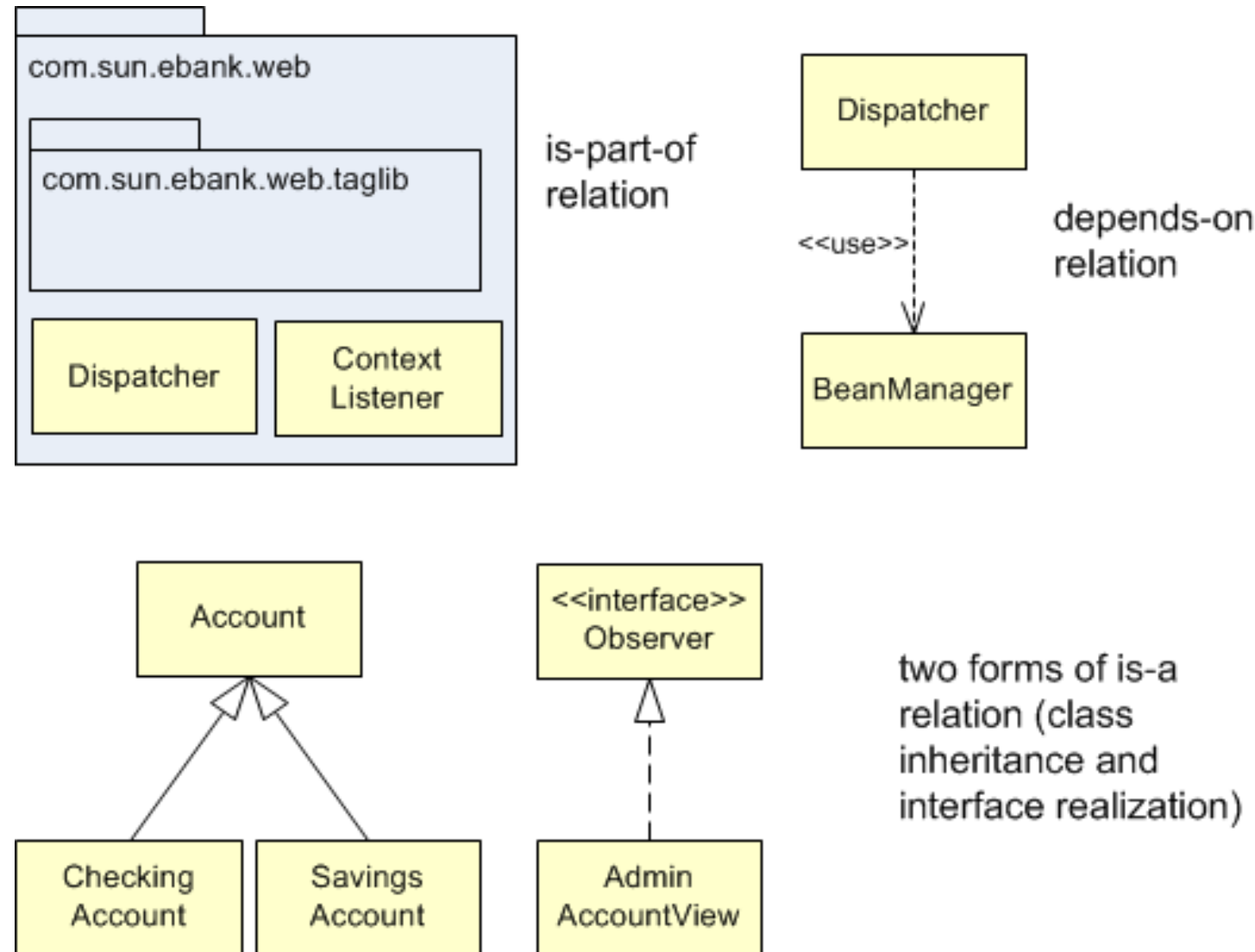
UML in module views – *elements*



UML doesn't use colors, but it's common to color symbols

Module views in UML are usually UML class diagrams

UML in module views – *relations*



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Module views

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Module decomposition view

- Shows the “is part of” relation between modules
 - In other words, it shows the decomposition of a module into submodules
- Module decomposition criteria include:
 - Improve modifiability
 - Build versus buy
 - Developers' skills

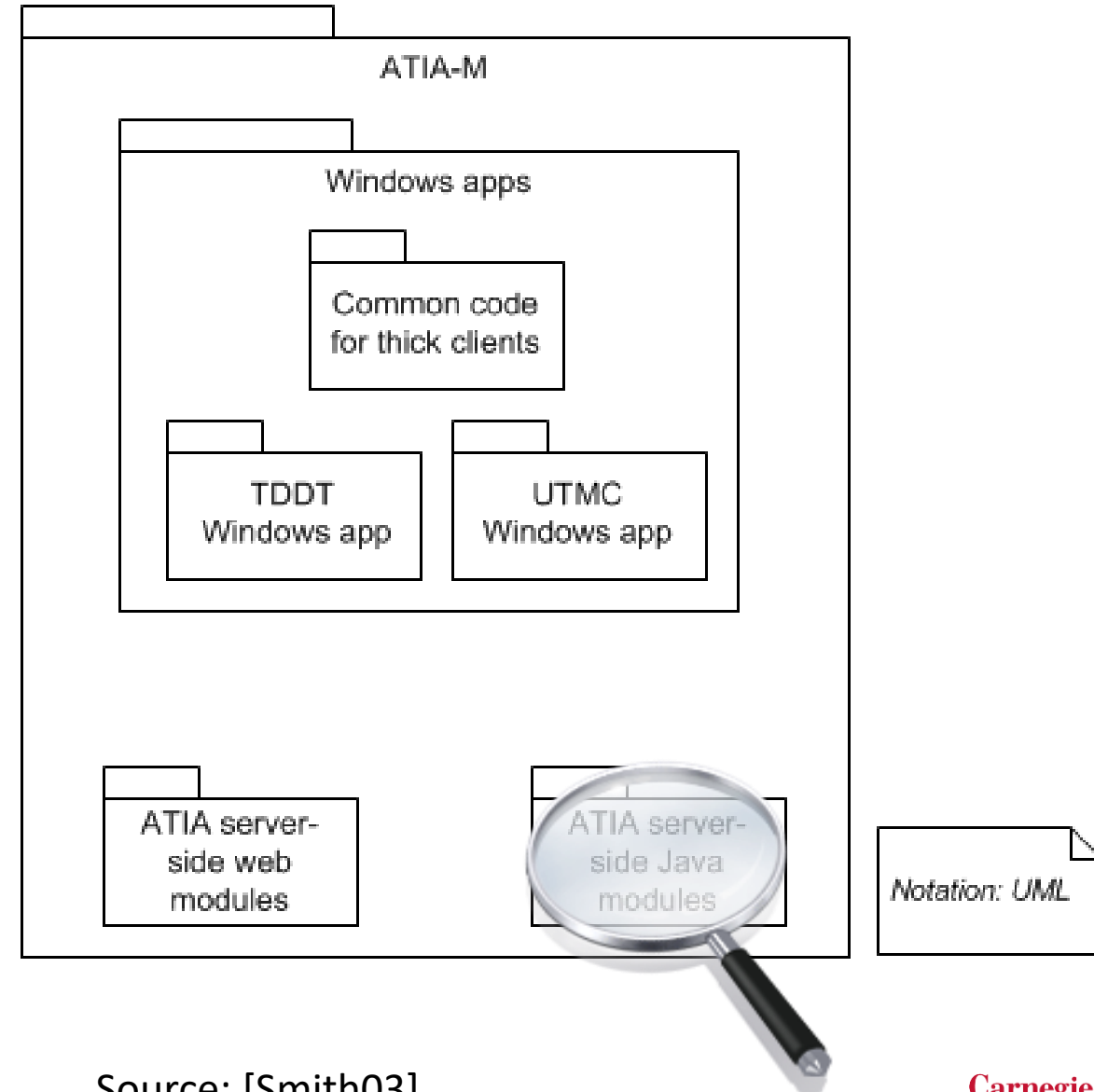
*Tactics we studied that affect
decomposition:*

- Split module
- Redistribute responsibilities

Module decomposition view – example 1

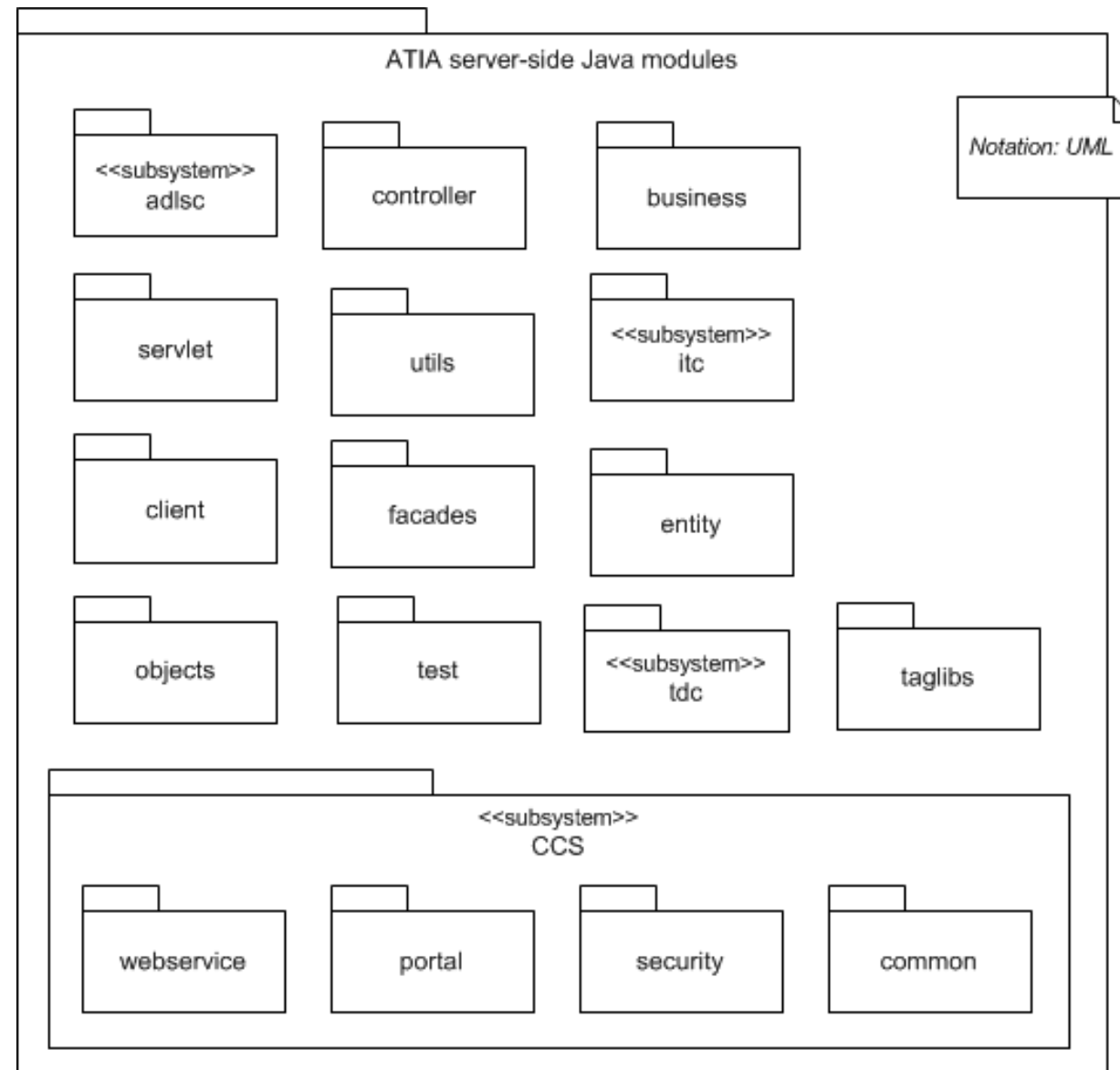
Disclaimer about examples:

- *no need to discuss or understand the design*
- *real and hence not perfect*
- *not the latest version*
- *diagram can't tell everything*



Source: [Smith03]

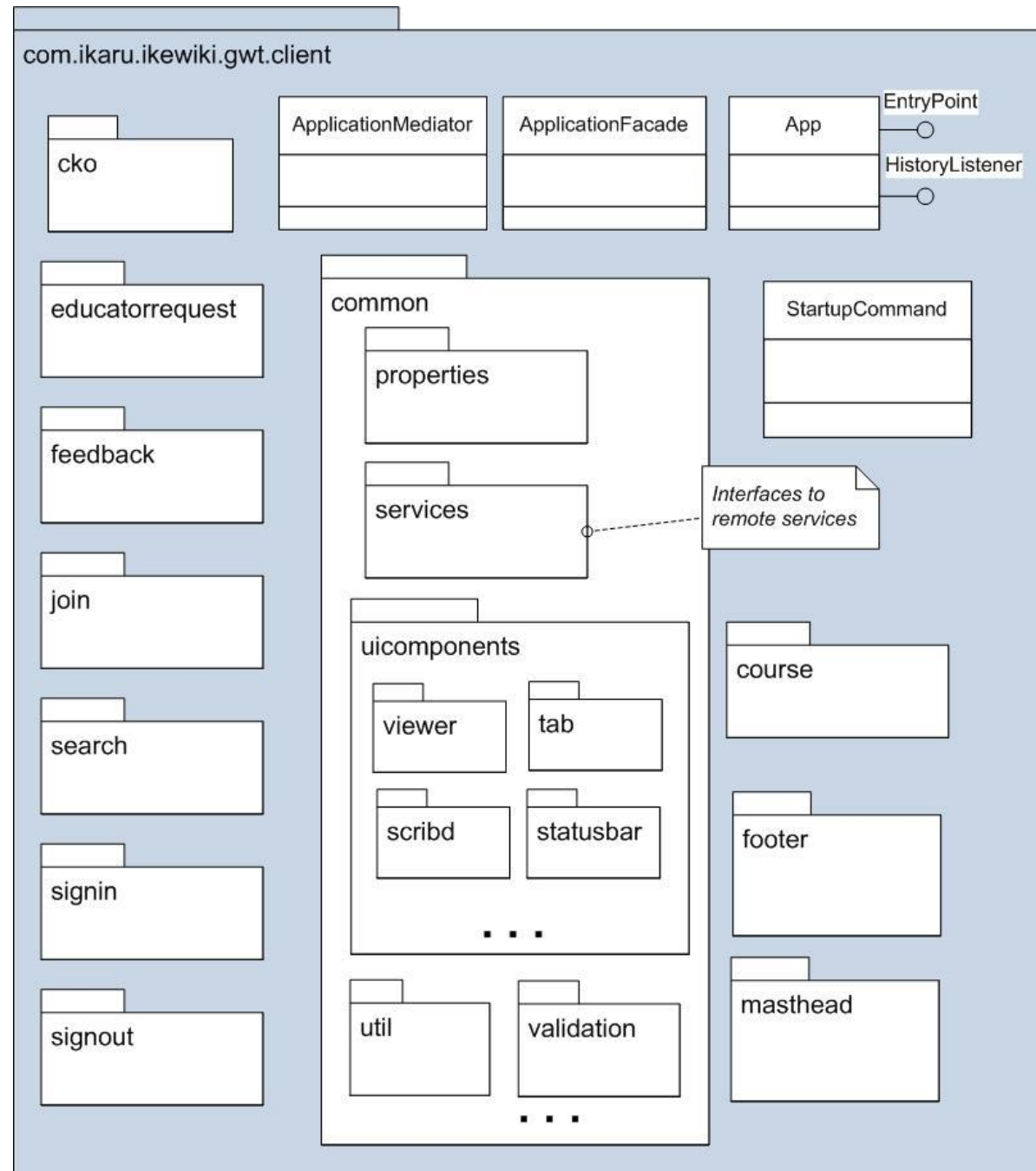
Module decomposition view – example 2



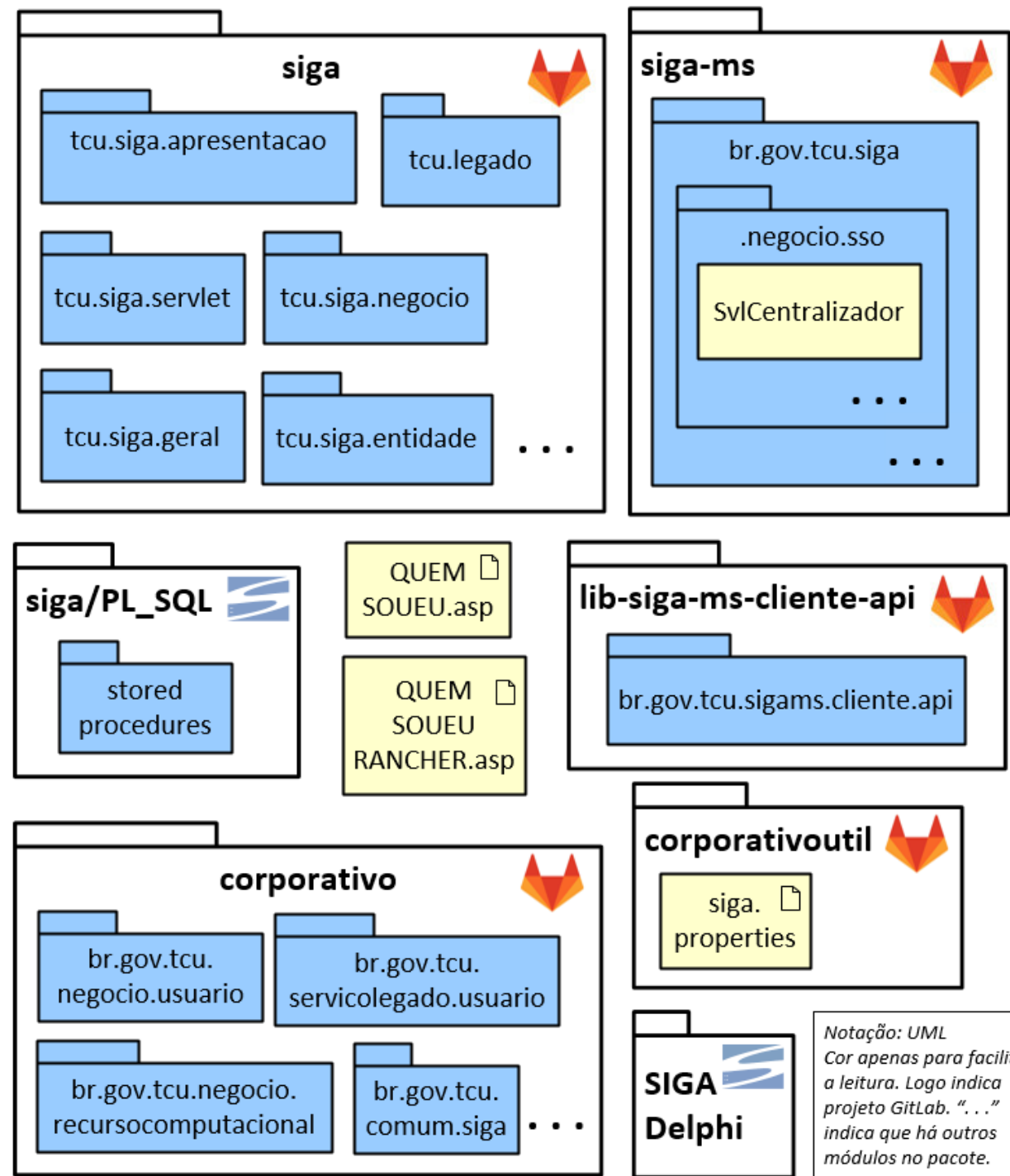
Source: [Smith03]

Module decomposition view – example 3

Key: UML
Color used to enhance readability only
"..." indicates there are other elements in the package
Dependencies not shown



Module decomposition view – example 4





Questions?

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